

Decision Trees

Intuition, Mathematics, and Python Implementation

Machine Learning Series | CodeWithHarry

Algorithm Intuition

Understanding how machines make choices

What is a Decision Tree?



Human-Like Reasoning

Decision Trees mimic how humans make decisions through a series of "Yes/No" questions. It is a non-parametric supervised learning method.

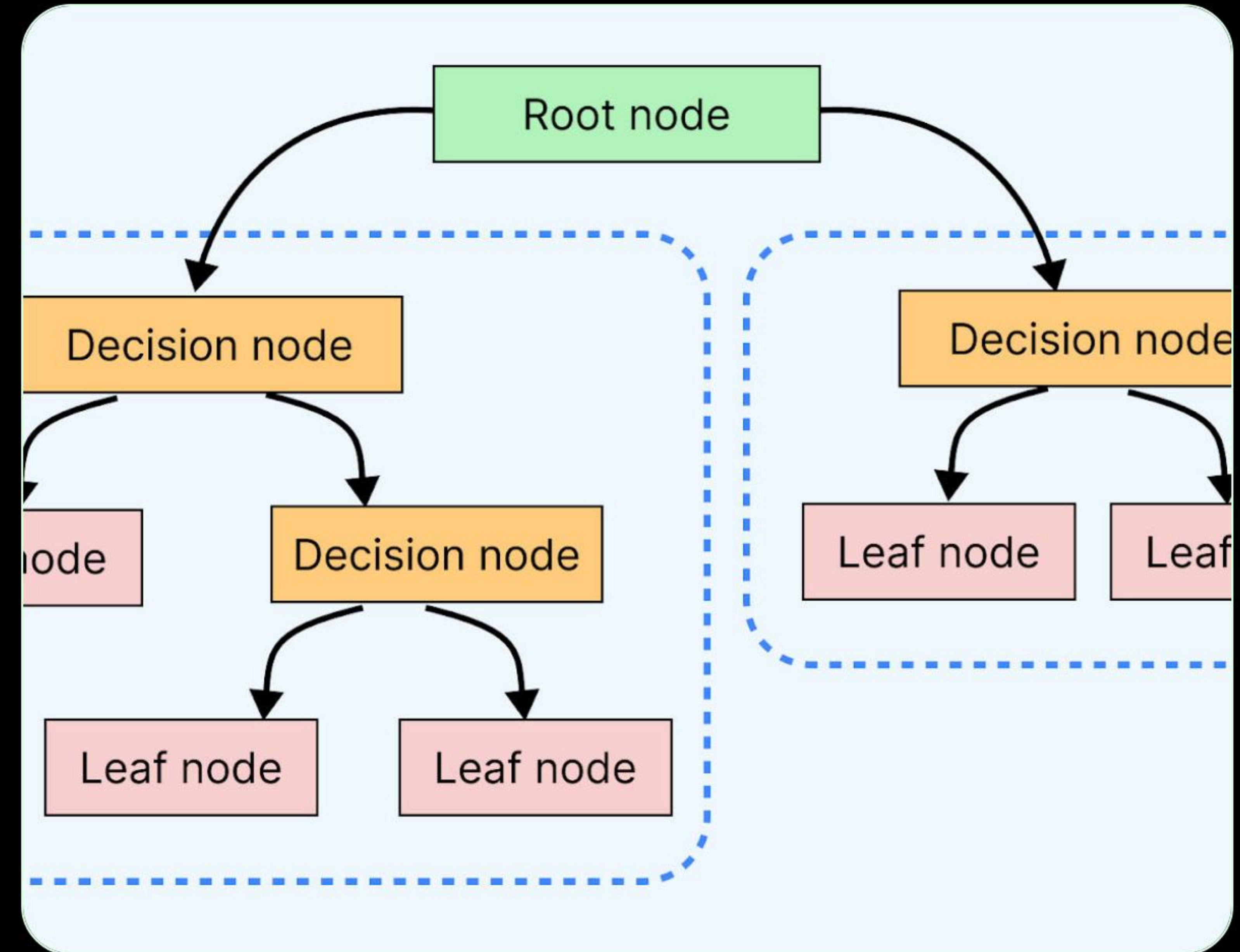


Hierarchical Structure

It breaks down a complex dataset into smaller subsets while at the same time an associated decision tree is incrementally developed.

Anatomy of the Tree

- **Root Node:** The starting point representing the entire population.
- **Decision Nodes:** Sub-nodes that split into further branches based on a test.
- **Leaf/Terminal Nodes:** The final nodes that do not split; these contain the predictions.
- **Branch:** The outcome of a decision node test.



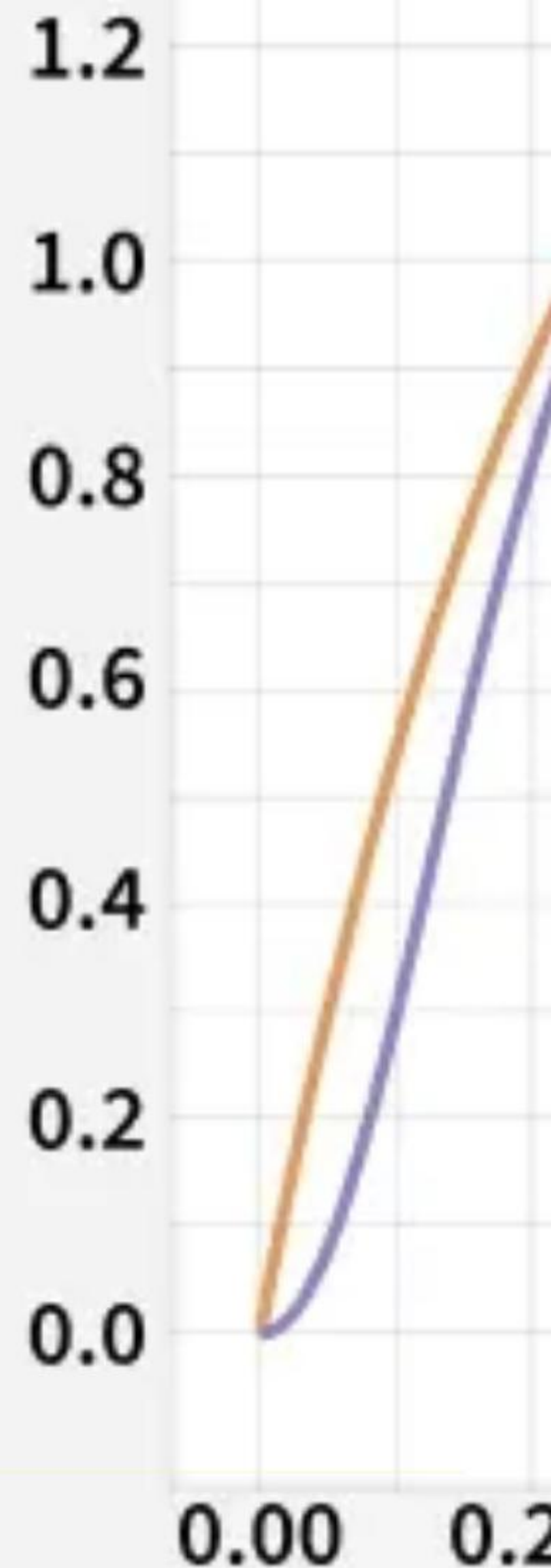
Entropy: Measuring Chaos

Entropy measures the "purity" or "disorder" of a node. A node is pure if all instances belong to the same class.

$$E(S) = - \sum_{i=1}^c p_i \log_2 (p_i)$$

Entropy is 0 when the node is perfectly pure and 1 when classes are distributed equally (50/50 split).

Entropy vs Gini Impurity



Information Gain

Gain

Max is Best

Logic of Splitting

Information Gain is the reduction in entropy achieved by partitioning data according to a specific feature.

We choose the feature that provides the highest Information Gain to split the node first.

$$IG = E(\text{parent}) - \sum w \cdot E(\text{child})$$

Gini Impurity

Computationally Faster

Gini impurity is faster to compute than entropy because it doesn't involve logarithmic calculations.

$$\text{Gini} = 1 - \sum_{i=1}^n p_i^2$$

Range: 0 to 0.5

Unlike Entropy (0 to 1), Gini index ranges from 0 to 0.5 for binary classification. Scikit-learn uses Gini as the default criterion.

Case Study: Weather Dataset

A classic example: Predicting if a student will play tennis based on the weather conditions.

OUTLOOK	TEMP	HUMIDITY	WIND	PLAY?
Sunny	Hot	High	Weak	No
Overcast	Hot	High	Weak	Yes
Rainy	Mild	High	Strong	No
Rainy	Cool	Normal	Weak	Yes

Splitting on Features



Hot



Lighting



Rain



Cold



Hail



Wind



Lighting
with sun



Sleet

1. Analyze Attributes

Calculate Entropy for Outlook, Humidity, and Wind.



2. Calculate Gain

Subtract weighted average entropy of children from parent entropy.



3. Choose Best

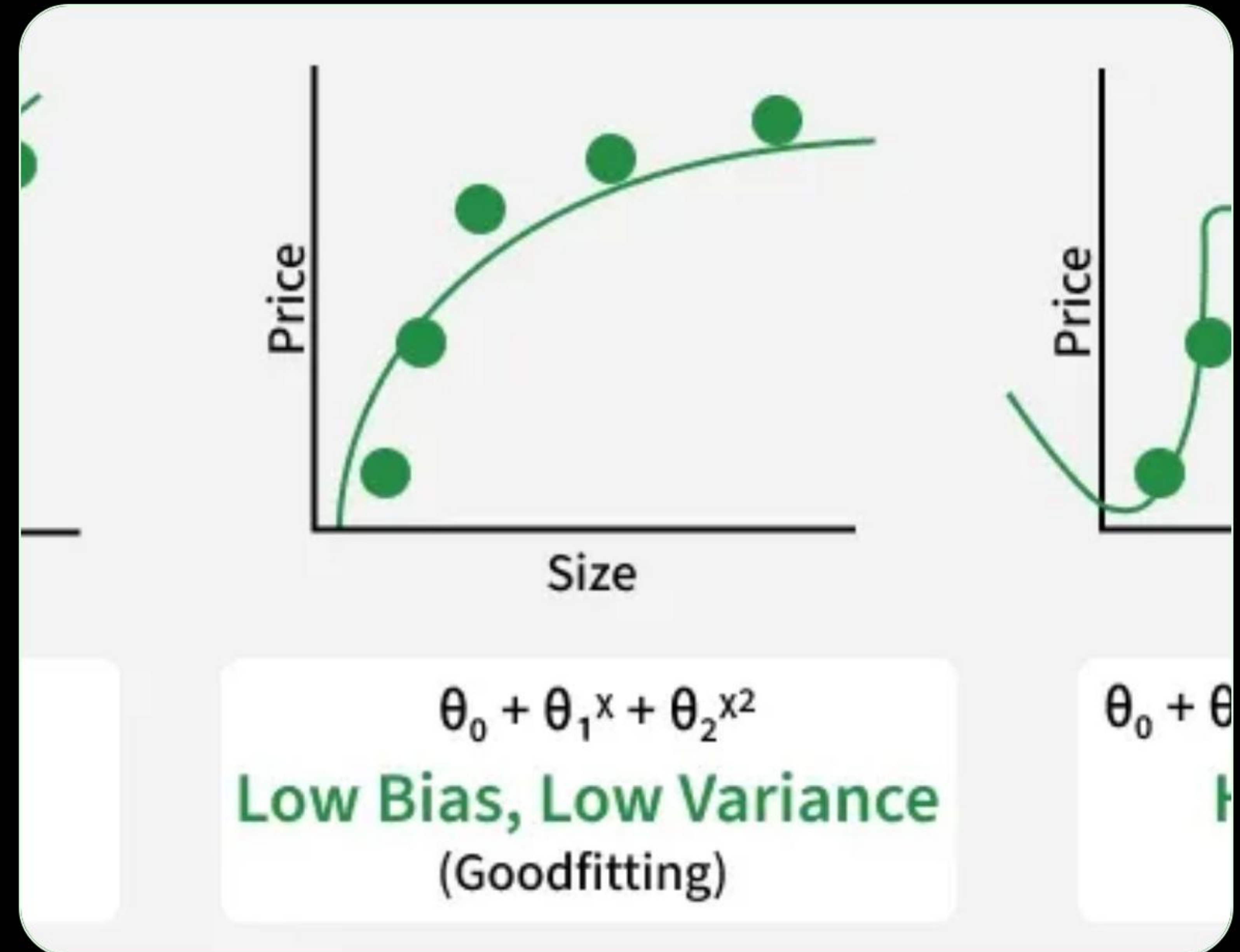
Select the attribute with the highest gain as the Root Node.

The Overfitting Problem

Growing Too Deep

A tree that is allowed to grow until every leaf is pure will likely "memorize" the training data, leading to poor performance on new data.

- High Variance.
- Sensitive to noise.
- Complex decision boundaries.



Tree Regularization

Pre-Pruning

Set a `max_depth` or `min_samples_split` to stop the tree from growing too large early on.

Post-Pruning

Allow the tree to grow fully and then remove nodes that provide little information using Cost Complexity Pruning.

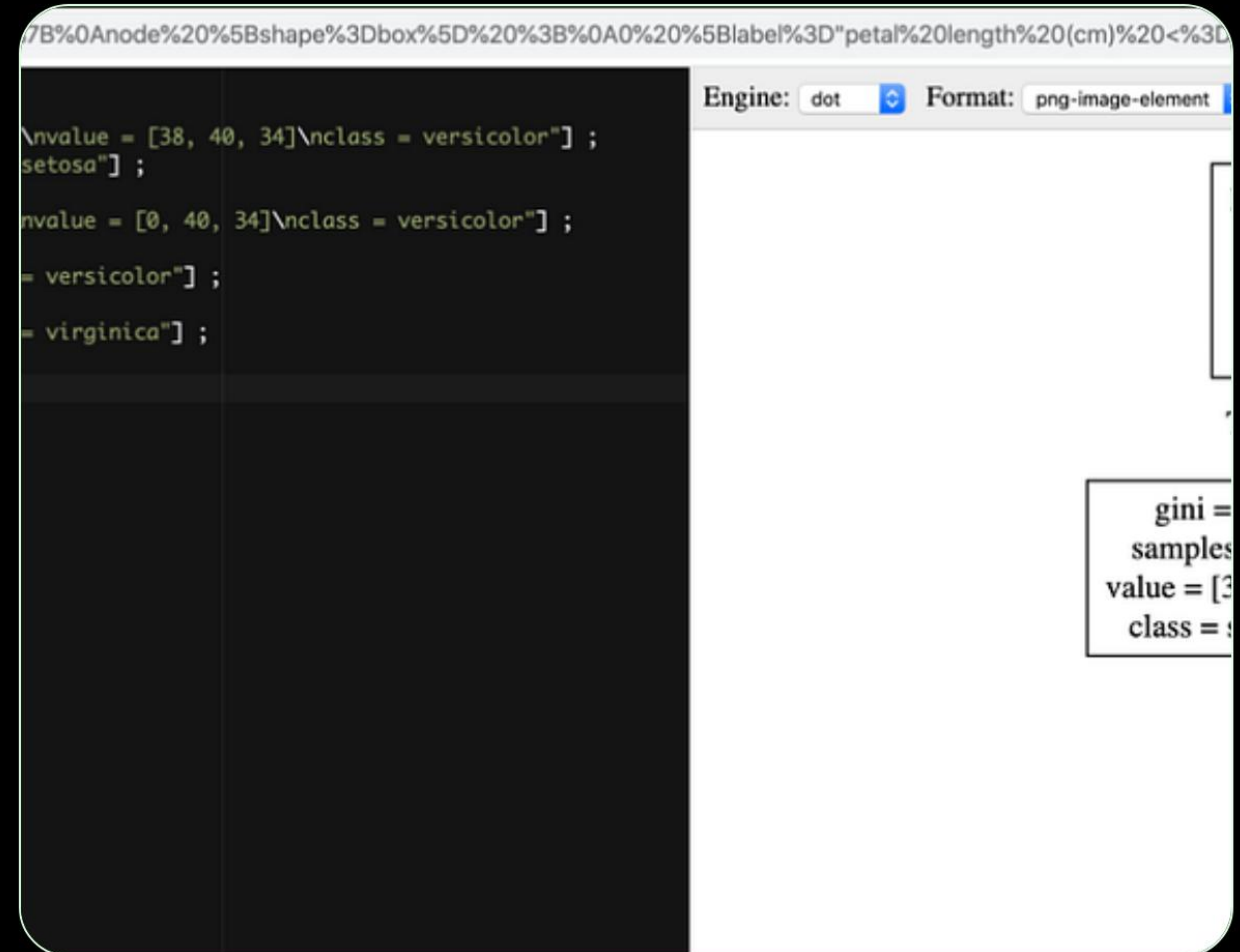
Python Implementation

Using Scikit-Learn

Implementation is straightforward with the `DecisionTreeClassifier`. You can easily visualize the resulting tree using `plot_tree`.

Common parameters:

- `criterion`: 'gini' or 'entropy'.
- `max_depth`: Controls depth.

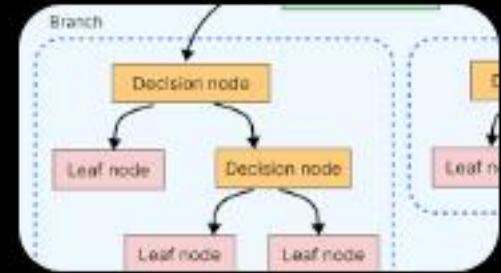


Any Questions?

Thank you for your attention!

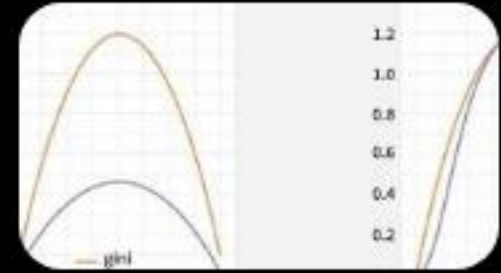
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Image Sources



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Source: www.upgrad.com



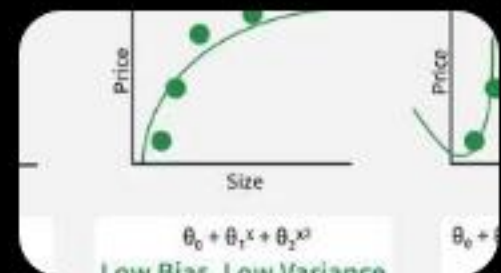
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